MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 01 (PROBABILITY)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 14: Probability** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 1 VSAQ question (2 marks). Section C has 3 SA questions (3 marks each).
3. Section D has 1 Applied Case Study (4 marks). Show all working clearly.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. Which of the following cannot be the probability of an event? **[1]**

Options: (a) $\frac{2}{3}$ (b) -1.5 (c) 15% (d) 0.7

2. Two unbiased coins are tossed simultaneously. The probability of getting at most one head is: **[1]**

Options: (a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1

3. A card is drawn from a well-shuffled deck of 52 cards. The probability of getting a red face card is: **[1]**

Options: (a) $\frac{3}{13}$ (b) $\frac{3}{26}$ (c) $\frac{1}{26}$ (d) $\frac{3}{52}$

4. A bag contains 3 red balls, 5 black balls, and 4 white balls. A ball is drawn at random from the bag. The probability of not getting a black ball is: **[1]**

Options: (a) $\frac{5}{12}$ (b) $\frac{7}{12}$ (c) $\frac{1}{3}$ (d) $\frac{1}{4}$

5. **Assertion (A):** The probability of an impossible event is 0.
Reason (R): The probability $P(E)$ of any event E always lies between 0 and 1, i.e., $0 \leq P(E) \leq 1$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [2 Marks]

6. Cards each marked with one of the numbers 4, 5, 6, ..., 20 are placed in a box and mixed thoroughly. One card is drawn at random. What is the probability of getting an even number? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [9 Marks]

7. Two unbiased dice are thrown simultaneously. Find the probability of getting: (i) a sum of 8 (ii) a sum greater than 9 (iii) an even sum. **[3]**

8. All the red face cards are removed from a pack of 52 cards. The remaining cards are well shuffled and a card is drawn at random. Find the probability of getting: (i) a red card (ii) a queen (iii) an ace of hearts. **[3]**

9. A jar contains 24 marbles, some are green and others are blue. If a marble is drawn at random from the jar, the probability that it is green is $\frac{2}{3}$. (i) Find the number of green marbles in the jar. (ii) Find the number of blue marbles in the jar. (iii) What is the probability of drawing a blue marble? **[3]**

SECTION D: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 10: Case Study Analysis

A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 marked on a circular wheel, as shown in the carnival stall. Each of the 8 outcomes is equally likely. Avni decides to play the game by paying an entry token.

Based on the above scenario, answer the following:

(i) What is the probability that the arrow points at an odd number? **[1]**

(ii) What is the probability that the arrow points at a prime number? **[1]**

(iii) What is the probability that the arrow points at a number greater than 2? **[1]**

(iv) What is the probability that the arrow points at a number less than 9? **[1]**

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